INCH-POUND

MIL-DTL-24640/13D
22 August 2002
SUPERSEDING
MIL-C-24640/13C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-PAIR, AWG 26, TYPE 2XSX0

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Non-Watertight)

First - 19-strand, tin-coated copper conductor, AWG size 26.
Second - Component wires shall be MIL-W-81044/12, except identification shall be in accordance with the codes and methods of MIL-DTL-24640. Testing shall be in accordance with the component wire ratings and requirements listed on this slash sheet. Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.
Third - Two conductors, one black and one white, twisted together. The pitch shall be 3/4 $\pm$ 1/8 inch.
Fourth - Binder tape applied helically with overlap, at manufacturer's option.
Fifth - Braided shield of # 38 AWG tin coated copper strands. Braid coverage shall be no less than 90 percent determined in accordance with WC-27500.
Sixth - Shield insulation of two polyester tapes or a polyester tape and extruded jacket. Identification codes applied using Method 2.
Seventh - Four shielded pairs shall be cabled together. Cabling sequence shall be consecutive. Fillers may be used as necessary, to form a round cable.
Eighth - Binder tape applied helically with overlap.
Ninth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Tenth - Binder tape applied helically with overlap, at manufacturer's option.
Eleventh - Crosslinked polyolefin jacket. (See Table I for thickness).

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TABLE I. Construction details.

PIN M24640/13	Type number	Number of pairs	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	2XSXO-4	4	0.243	0.038	0.333	0.359	101.0

REQUIREMENTS:

Qualification required.
Construction details: See Table I.

Component wire: The component wire shall be per MIL-W-81044 and shall be tested in accordance with the following wire ratings and additional requirements.
Temperature rating: 150°C (302°F) maximum conductor temperature
Voltage rating: 600 volts (rms) at sea level.
Accelerated aging: Oven temperature, 300 ± 2°C (572 ± 3.6°F) for 6 hours.
Blocking: 150 ± 2°C (302 ± 3.6°F)
Flammability: 30 seconds (max); 3.0 inches (max); no flaming of tissue paper.
Humidity resistance: 5000 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test:
Primary insulation (when tested in lieu of spark test):
6.0 kilovolts (peak) 100 percent test.
Finished wire: 8.0 kilovolts (peak), or 3 kHz spark test
5.5KV, 100 percent test.
Insulation resistance: 5000 megohms for 1000 feet (min).
Life cycle: Oven temperature, 200 ± 2°C (392 ± 3.6°F) for 168 hours.
Physical properties of primary insulation:
Tensile strength, 2500 lb/in^2 (min).
Elongation, 150 percent (min).
Shrinkage: 0.125 inch maximum at 300 ± 2°C (572 ± 3.6°F).
Spark test primary insulation: 1500 volts (rms), 60 Hz or 3 kHz, 100 percent test.
Surface resistance: 500 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.060 inch.
Wet dielectric test: 2500 volts (rms).
Wrap test: Mandrel wrap test required.

EXAMINATION AND TESTS:

Basic electrical: Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum.....44.24

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Voltage withstand minimum - volts, root mean square,
conductor and shield 2000
Conductor to conductor and shield .....Not applicable
Shield to shield.....200
Component shield to overall shield 500
Insulation resistance – megohms-1000 feet, minimum
Conductor to shield.....2500
Shield to shield (component shield only) 100
Shield to shield (component shield to overall shield).....100
Conductor and shield continuity .....No failure
Jacket flaws .....No failure

Group A:
Visual and dimensional.....No failure
Capacitance:
Mutual capacitance - at 1 MHz, picofarads/foot (pF/ft),
maximum .....Not applicable
Capacitance unbalance - percent, maximum .....Not applicable
Characteristic impedance - at 1 MHz, ohms.....Not applicable
Watertightness (see Table IX of MIL-DTL-24640 for
limits of water leakage).....Not applicable
Weight .....No failure

Group B:
Cold bending.....No failure
Drip .....Not applicable
Crosslink proof test (percent maximum)
Jacket 50
Tear - pounds per inch thickness, minimum.....35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum .....Not applicable
Gas flame (1 hour) .....Not applicable

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - cycles, minimum .....Not applicable
Permanence of printing (jacket) - cycles, minimum.....125
Shield - conformance to material, construction and coverage .....No failure

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Heat distortion - percent of unaged, maximum 30
Cable sealant removability.....Not applicable
Shrinkage .....No failure
Mutual inductance - dB/1000 feet, maximum .....Not applicable
Surface transfer impedance
    Milliohms per meter, maximum 700
    EMP response, dB, minimum 60
Pulse response time - ms/500 feet
    Rise time, maximum .....Not applicable
    Decay time, maximum .....Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Flame propagation .....No failure
Aging and compatibility (cable) .....No failure
Acid gas equivalent - percent, maximum
    Jacket 2
    Fillers 2
    Insulation 18
Halogen content - percent, maximum
    Jacket 0.2
    Fillers 0.2
Hot oil immersion
    Tensile strength - percent of unaged, minimum 50
    Elongation - percent of unaged, minimum 50
Smoke index, maximum
    Jacket 25
    Fillers 45
    Insulation 10
Toxicity index, maximum
    Jacket 5
    Fillers 5
    Insulation 1.5
Durometer (jacket) - (type A) hardness, minimum.....80
Weathering (jacket).....No failure
Fungus resistance.....No failure
Abrasion resistance.....No failure
Gas Flame (3 hour) .....Not applicable
Component wire.....No failure

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Unit ordering length: 500 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-013)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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